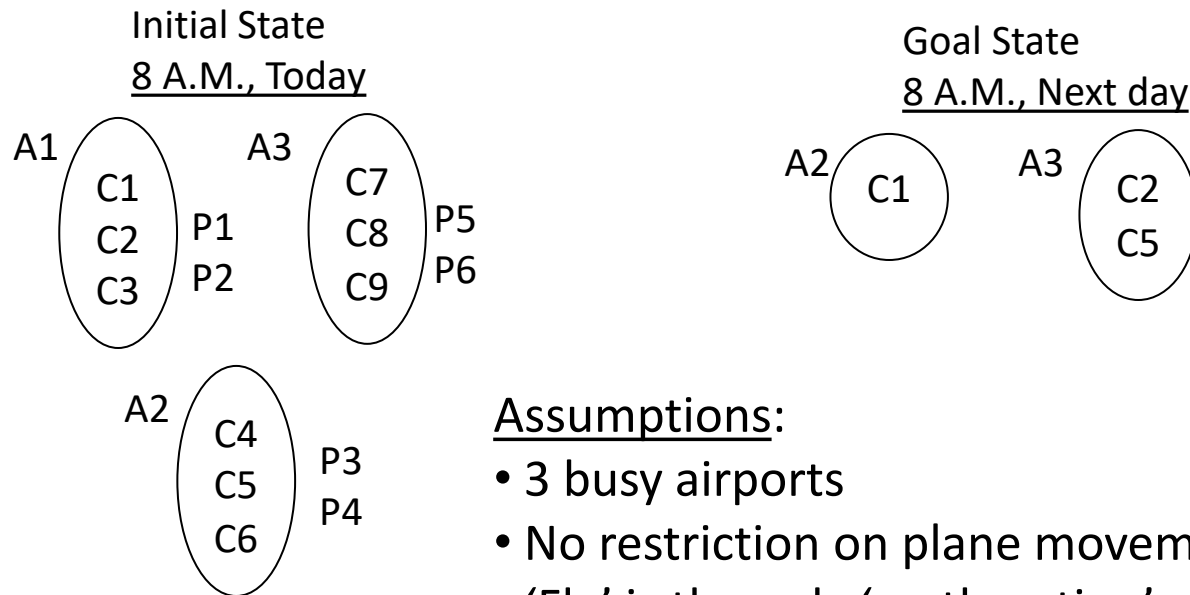


Chapter 4. Planning Problems

Topic 4.1. Air-Cargo Transportation Problem as a Planning Problem

A) Representation of the problem

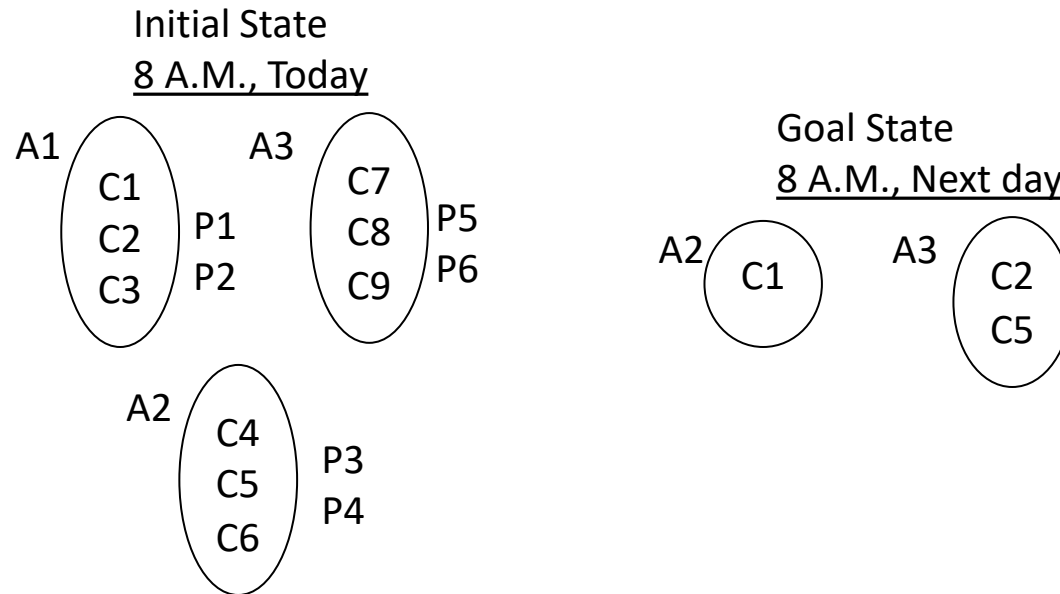
a) Schematic representation



Assumptions:

- 3 busy airports
- No restriction on plane movement
- 'Fly' is the only 'costly action'
- Any plane can reach another port directly or via the 3rd by the given time

b) Representation in Planning Domain Definition Language (PDDL)



Initial State (IS):

$\text{Airport}(A1) \wedge \text{Airport}(A2) \wedge \text{Airport}(A3) \wedge \text{Cargo}(C1) \wedge \text{Cargo}(C2) \wedge \dots$
 $\wedge \text{Cargo}(C9) \wedge \text{Plane}(P1) \wedge \text{Plane}(P2) \wedge \dots \wedge \text{Plane}(P6) \wedge \text{At}(A1, C1) \wedge$
 $\text{At}(A1, C2) \wedge \dots \wedge \text{At}(A3, C9) \wedge \text{At}(A1, P1) \wedge \text{At}(A1, P2) \wedge \dots \wedge \text{At}(A3, P6)$

Goal State (GS):

$\text{At}(A2, C1) \wedge \text{At}(A3, C2) \wedge \text{At}(A3, C5) \wedge \text{Airport}(A2) \wedge \text{Airport}(A3) \wedge$
 $\text{Cargo}(C1) \wedge \text{Cargo}(C2) \wedge \text{Cargo}(C5)$

c) Action Schemas:

i) Action: Load(c, p, a)

Precondition: $\text{Cargo}(c) \wedge \text{Plane}(p) \wedge \text{Airport}(a) \wedge \text{At}(a, c) \wedge \text{At}(a, p)$

Effect: $\neg \text{At}(a, c) \wedge \text{In}(p, c)$

ii) Action: Unload(c, p, a)

Precondition: $\text{Cargo}(c) \wedge \text{Plane}(p) \wedge \text{Airport}(a) \wedge \text{At}(a, p) \wedge \text{In}(p, c)$

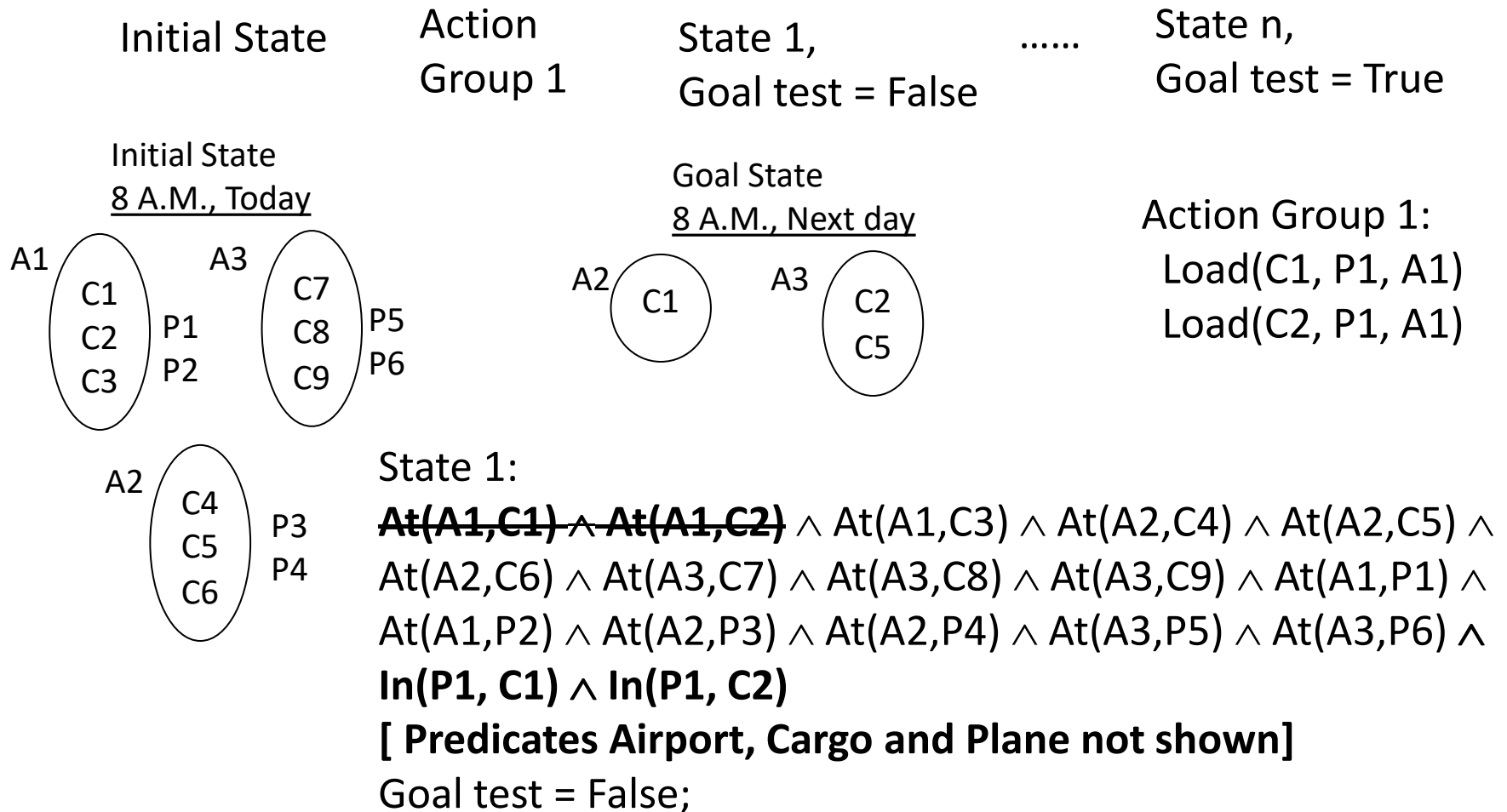
Effect: $\text{At}(a, c) \wedge \neg \text{In}(p, c)$

iii) Action: Fly(p, from, to)

Precondition: $\text{Plane}(p) \wedge \text{Airport}(\text{from}) \wedge \text{Airport}(\text{to}) \wedge \text{At}(\text{from}, p)$

Effect: $\neg \text{At}(\text{from}, p) \wedge \text{At}(\text{to}, p)$

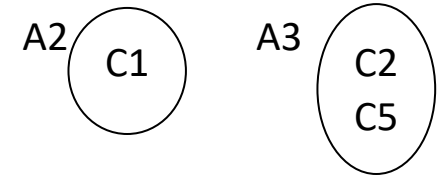
B) Construction of a plan through Forward State Space Search



State :

~~$\text{At}(A1, C1) \wedge \text{At}(A1, C2)$~~ $\wedge \text{At}(A1, C3) \wedge \text{At}(A2, C4) \wedge \text{At}(A2, C5) \wedge$
 $\text{At}(A2, C6) \wedge \text{At}(A3, C7) \wedge \text{At}(A3, C8) \wedge \text{At}(A3, C9) \wedge \text{At}(A1, P1) \wedge$
 $\text{At}(A1, P2) \wedge \text{At}(A2, P3) \wedge \text{At}(A2, P4) \wedge \text{At}(A3, P5) \wedge \text{At}(A3, P6) \wedge$
 $\text{In}(P1, C1) \wedge \text{In}(P1, C2)$

Goal State
8 A.M., Next day



Goal Test =

Action Group2:

Fly(P1, A1, A2)

Action Group 3:

Unload(C1, P1, A2)

Load(C5, P1, A2)

Action Group 4:

Fly(P1, A2, A3)

Action Group 5:

Unload(C2, P1, A3)

Unload(C5, P1, A3)

Plan: Action Group1, Action Group 2, ... , Action Group 5; Optimal?